Determining Bone Health Utilizing Visible Light

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Introduction

The goal of this project is to measure post-mortem bone porosity using light to develop a forensic test. Knowing that bones are porous, the consequential thought was that light would be allowed through the bone at different levels based on porosity.

Anatomy, Physiology, and Forensic Analysis of Bones

Bones are made of a mixture of collagen and calcium, which gives bone the necessary combination of flexibility and rigidity respectively. There are two types of bones: hard and heavy compact/cortical bone and porous spongy bone.[5]

Bones, once macerated, can be examined to create a biological profile. Maceration is a process of cleaning bones, consisting of placing the bone in a sealed container of water and allowing bacteria to eat the soft tissue. Examining bone composition at the visible and cellular level can differentiate between human and animal bones, and age through sutures, teeth, and eburnation.[4]

Durham University conducted a study in which whole bovine rib bones were boiled. Boiling a bone crystallizes apatite, a reflective mineral in bone. Boiling a bone also increases porosity, as the bone loses minerals, fats, and proteins such as collagen.[7]

Osteoporosis

In the United States alone, 2.1 million osteoporosis-related fractures are reported each year, resulting in $20.3 billion in healthcare costs.[1] Worldwide, there are more than 200 million cases of osteoporosis. [11] Osteoporosis occurs when spongy bone becomes more porous and cortical bone thins, making the bones less dense than normal. The decrease in density makes the bones more at risk of fracture.[8]

Density can be determined by measuring the grams of calcium in a bone segment using a special dual energy X-ray (DXA).[2] Central DXA tests are the most popular form of bone density tests and take measurements at your hip and spine, the most accurate spots. This test works on the premise that bone matter of different densities will either pass or block the transmitted X-rays. Depending on the level of absorption of the different X-rays, the test calculates a level of mineral content in the tested bone mass.

One of the units for bone mineral density (BMD) testing is T-score, which operates using standard deviation. T-score compares the results of the BMD test to the ideal bone mineral density of a healthy 25 year old male. Normal T-scores range from -1 to 1. Every standard deviation below zero indicates a higher risk of fracture. The T-score required for a diagnosis of osteoporosis is -2.5.[6]

Engineering Goals and Hypotheses

Engineering Goals	Hypotheses
Design and prototype a device that can measure the difference in LDR resistance between healthy bone and porous bone.	If a bone sample is exposed to a controlled source of light, then the amount of light transmitted through the bone will differ between healthy bone and porous bone.
Design and prototype a device that can measure the difference in bone porosity using visible light.	If porosity increases in a bone sample, then translucency will decrease.
	Human bone, when submitted to the same testing as bovine bone, will have different exact data points, but show a similar trend.

Design

Figure 1 displays the electronic design in Visuino, the visual programming platform for Arduino. The central element is the Arduino Uno. An Arduino is a microcontroller, which is a compact integrated circuit designed to perform specific tasks. Arduinos typically combine a processor, read-only memory, and input-output functions. The pulse generator element provides the timing for the number of samples per second. In this case, two samples are taken per second. The analog snapshot uses the timing from the pulse generator to capture the reading from the analog sensor. The analog sensor is the light dependent resistor (LDR), which is a sensor that varies its resistance based on the amount and concentration of light it detects. The analog snapshot passes the LDR reading as an output to the liquid crystal display (LCD). The elements on the LCD are customized for the outputs. The LM value in the LCD is a T-score estimate with a sample equation to show that it is possible for the Arduino to perform that math internally. The equation currently is $1/(R_{LDR})^2$.The Visuino Schematic (Figure 1) is downloaded and compiled by the Arduino IDE into code that can be executed on the microcontroller.

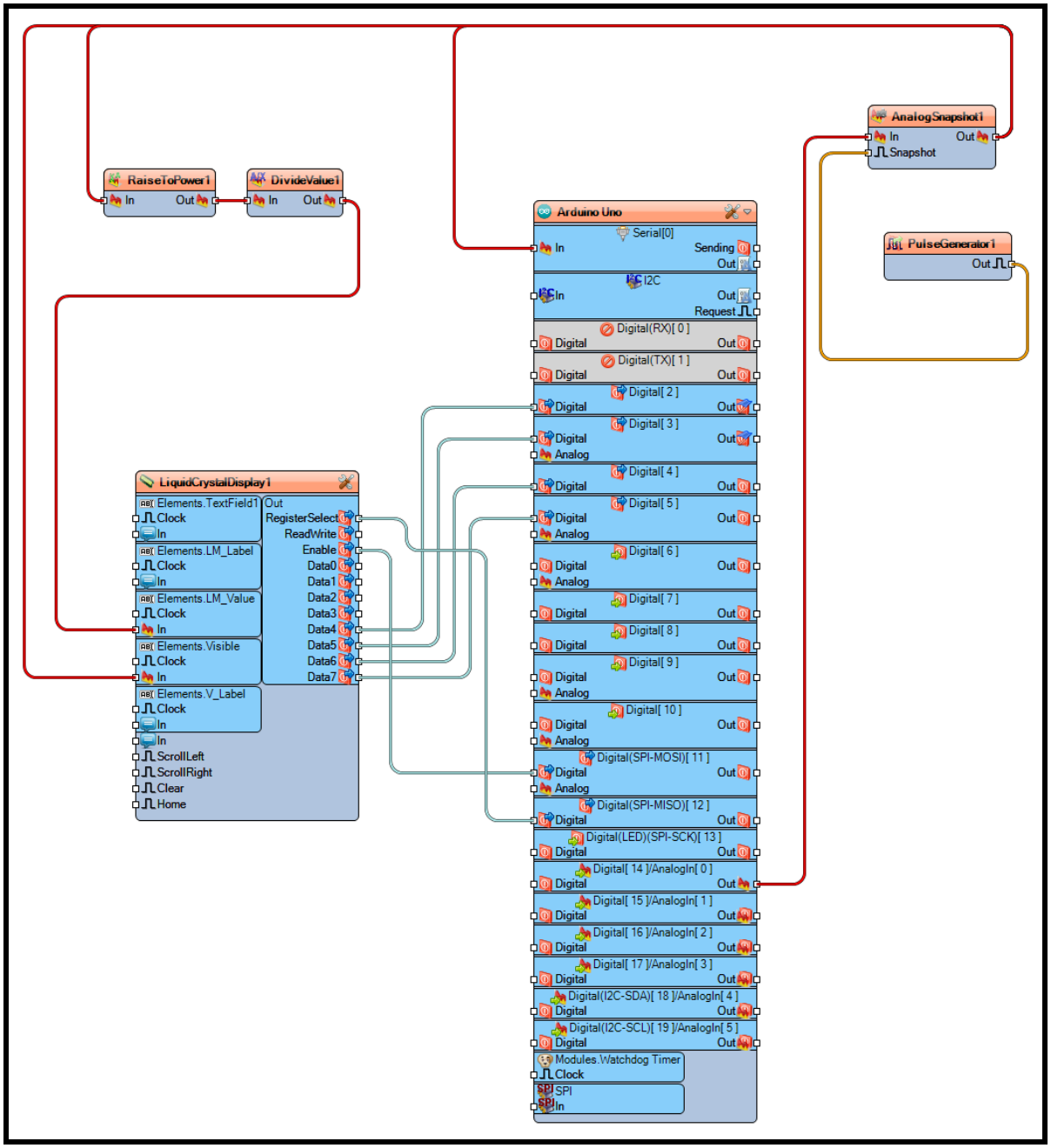


Figure 1: Visuino Sketch used to generate Arduino Code/Logical Design of the Electronics

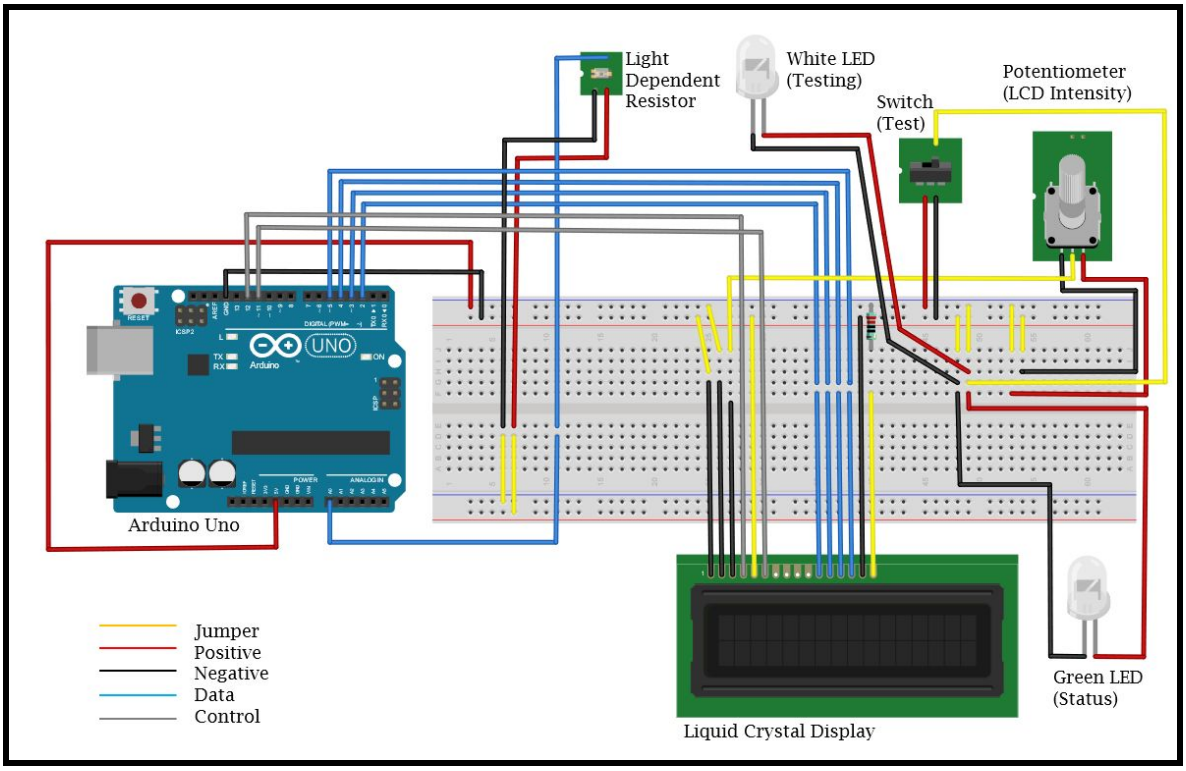


Figure 2: Electronics Pinout/Schematic of Circuitry Implemented

A prototype was constructed that provides a self-contained system with a sample chamber/stage, a light source, a light dependent resistor, and the corresponding electronics to collect the data from the light sensor, and display them in an LCD to the operator of the system. The schematic shown in Figure 2 represents the physical implementation of the logical design (Figure 1) in the prototype. Several components have been added, such as the switch, the LEDs, the potentiometer, and the breadboards through which everything connects. The switch turns on the white testing LED and the green status LED. On the actual device, there are four breadboards used, and items are jumpered from the main breadboard and Arduino. The resistor limits the voltage going to the backlight display of the LCD. The potentiometer controls the backlight intensity on the LCD so it can be adjusted according to the ambient light. There is no resistor on the testing LED, so it is always at full intensity while on, so it can burn out quickly, but can be just as easily replaced.

Light pollution would be when external light is allowed into the testing chamber, which would skew or possibly invalidate results. To prevent light pollution, all edges were lined with weather stripping, the entire device was painted black, and the LED was put in a grommet to concentrate the light. The LED and LDR are isolated in a compartment called the “stage” in which the bone sample is placed for data collection.

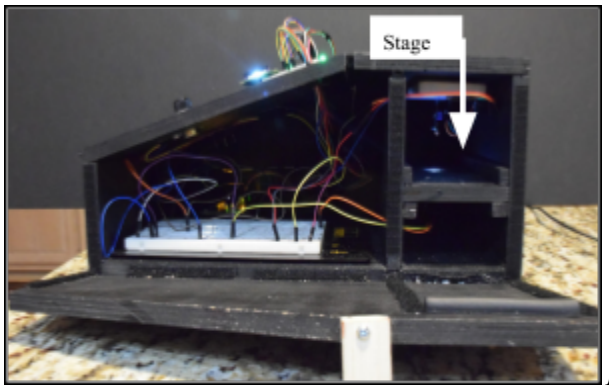


Figure 3: Image of Built Prototype with Electronics

Experimental Procedure

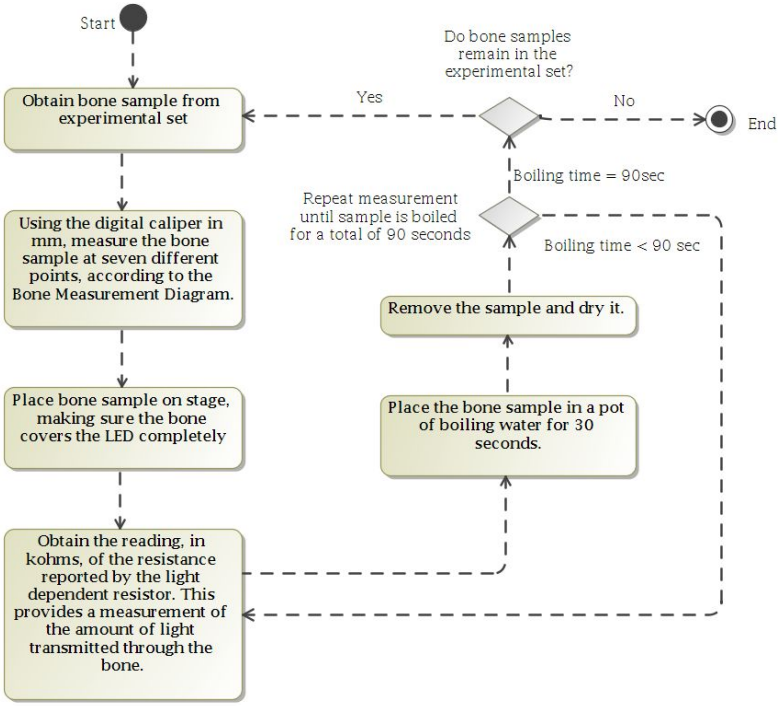


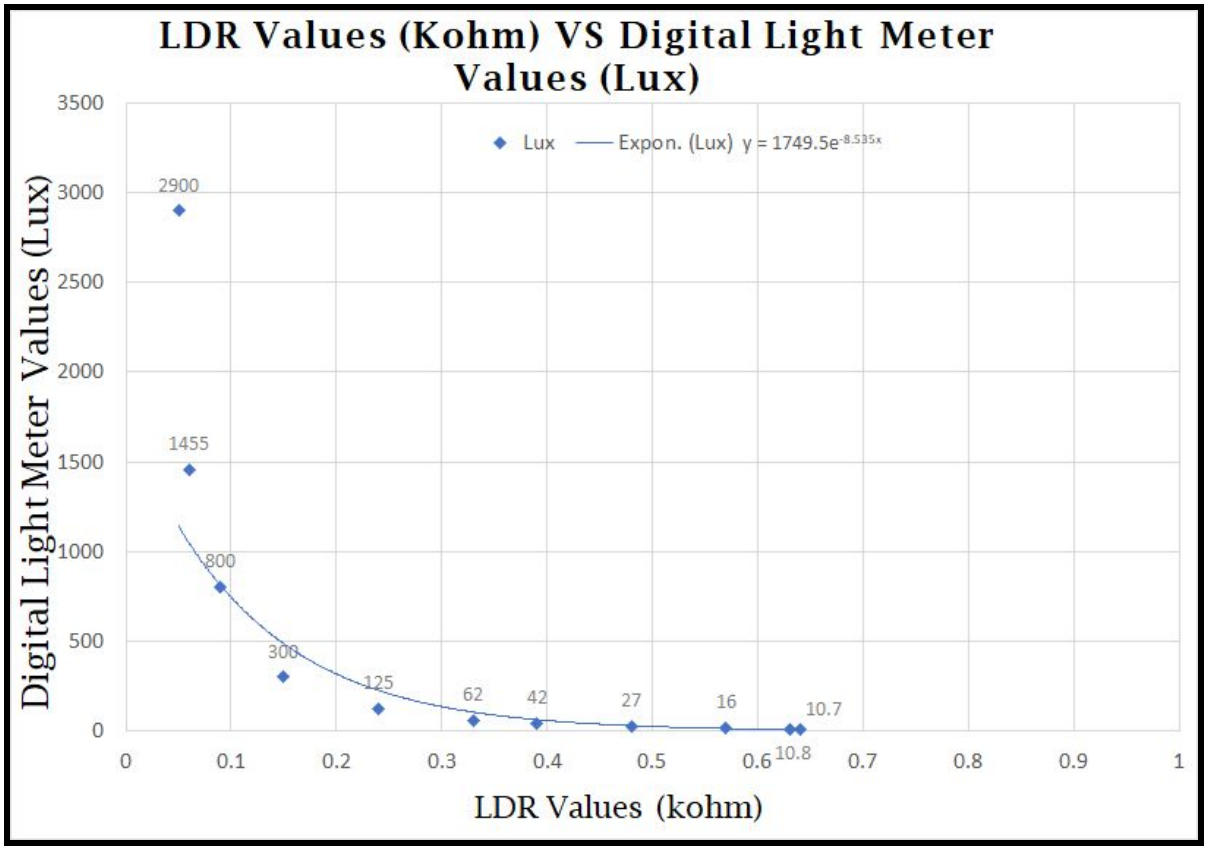
Figure 4: Experimental Procedure Flowchart

The experimental procedure displayed in Figure 4 is followed exactly for Experimental Sets 2 and 3, but is varied slightly in Experimental Set 1.

Seven measurements were taken of diaphyseal (a cylinder cut vertically) cross sections. Each of the 22 samples was measured in order to get baseline dimensions and classify the samples into one of two groups. One group, classified as "thin" was cut to an average thickness of 4.5mm. The other group, called "thick" was cut to approximately 11.37mm. There were 15 "thin" and 7 "thick" bone samples total. Five of the thin samples and all of the thick samples were part of Experimental Set 1 (Graph 2), and the other ten thin samples were part of Experimental Set 2 (Graph 3). The bone was boiled to manually induce osteoporosis-like conditions, as per the Durham University study.[7]

Data Results

Sensor Calibration



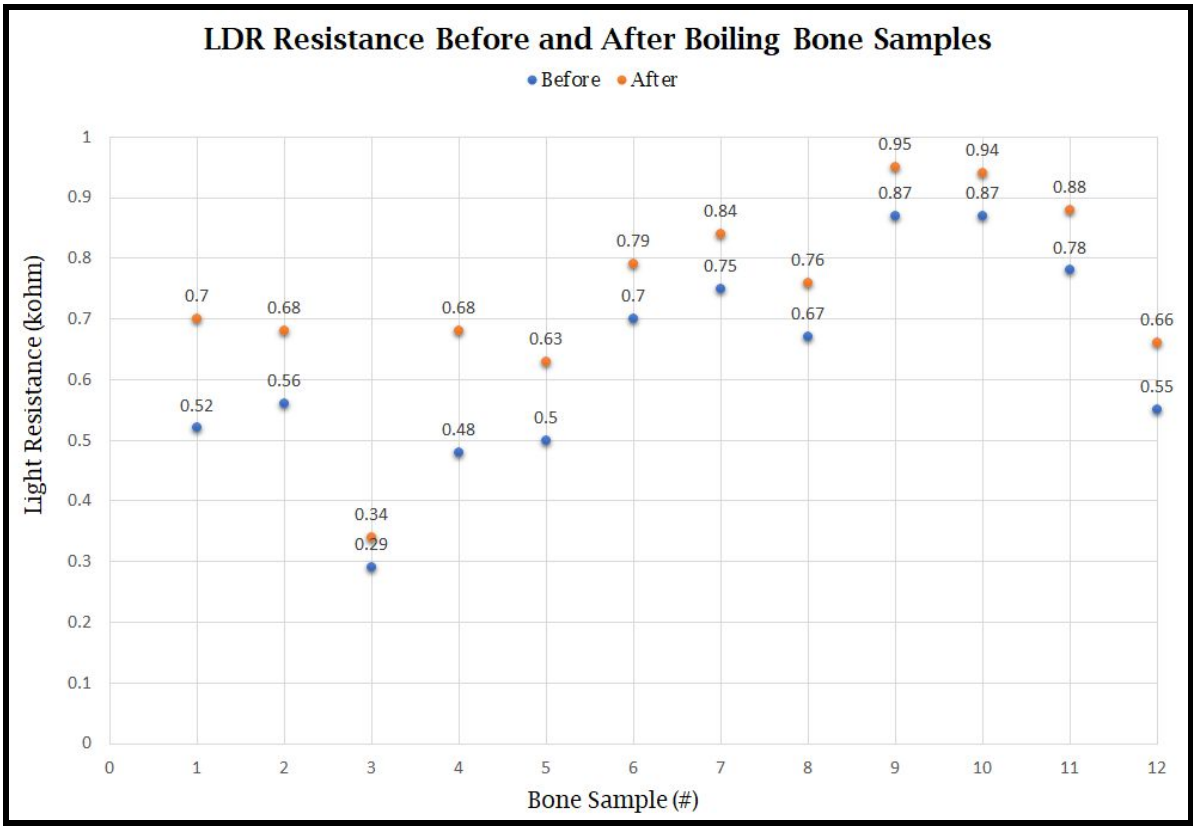
Graph 1: Sensor Calibration, LDR Values (kohm) VS Digital Light Meter Values (lux), and Exponential Equation

The SI unit for light measurement is lux which measures lumens per square meter. The LDR returns a value in ohms which is the measure of resistance. The LDR returns lower resistance values for brighter light which are higher lux values.[10]

To calibrate the LDR, simultaneous readings were taken from the LDR and a digital light meter held in close proximity to each other and in the same position relative to the visible light source. The light source was varied by filtering it with layers of individual sheets of white printer paper. The lux values from the digital light meter and the kohm values from the light dependent resistor were plotted against each other. Since light decay is an exponential function, an exponential regression automatically generated in Microsoft Excel was chosen to create the line of best fit.

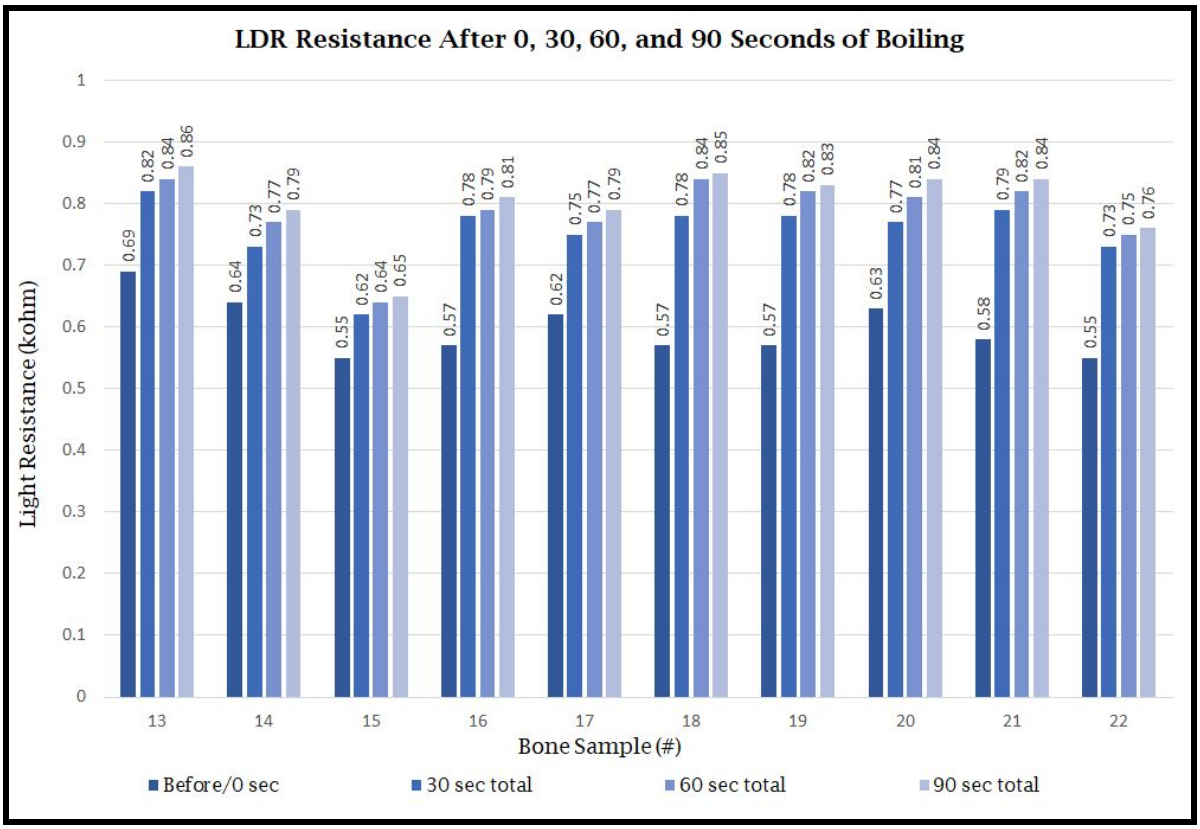
Bovine Experimentation

For this experiment, a total of 22 diaphyseal cross sections were taken from Texas Wagyu femur bones. Bovine bone is an acceptable substitute for human bone for forensic purposes.[12] All boiling occurred at a temperature of 100ºC, which was monitored using a digital cooking thermometer.



Graph 2: LDR Resistance (kohm) Before and After Boiling Bone Samples in Experimental Set 1

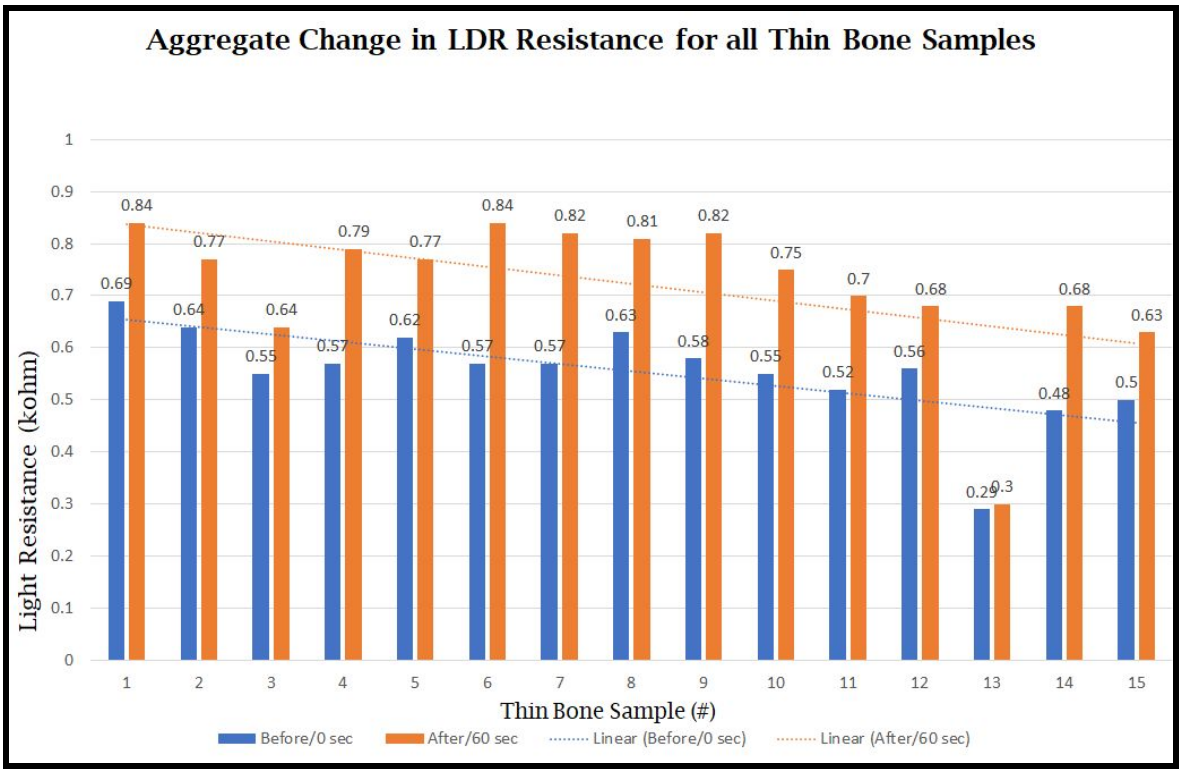
In Experimental Set 1, samples #1 through #5 are "thin" samples, and samples #6 through #12 are "thick". After boiling, samples presented a higher LDR resistance, meaning they are less translucent and more porous. The average LDR resistance for thin samples before boiling was 0.47 kohm; for thick samples, the resistance before boiling was 0.74 kohm. After boiling, the resistance measured when testing thin samples was 0.61 kohm and 0.83 kohm for thick samples. This translates into an average increase in LDR resistance of 29.8% for thin samples and 12.2% for thick samples.



Graph 3: LDR Resistance (kohm) After 0, 30, 60, and 90 Seconds of Boiling in Experimental Set 2

In Experimental Set 2, samples #13 through #22 are all thin (4.5 mm) bone samples. The data displays the increase in LDR resistance in 30 second increments for boiling. The most significant change occurred in the first 30 seconds of boiling. As boiling time increases, LDR resistance also continues to increase, but at a slower pace. This is likely due to lower amounts of proteins remaining in the bone at each interval, resulting in a smaller change as the residual protein evacuates the bone during the boiling process.

The average LDR resistance before boiling is 0.6 kohm, and, after 30 seconds of boiling, it increases to 0.76 kohm (an increase of 0.16 kohm). Average LDR Resistance after 60 seconds is 0.79 kohm and after another 30 seconds of boiling, increases to 0.8 kohm. There was an overall (from zero to ninety) increase of 0.2 kohm, or 33.3%.

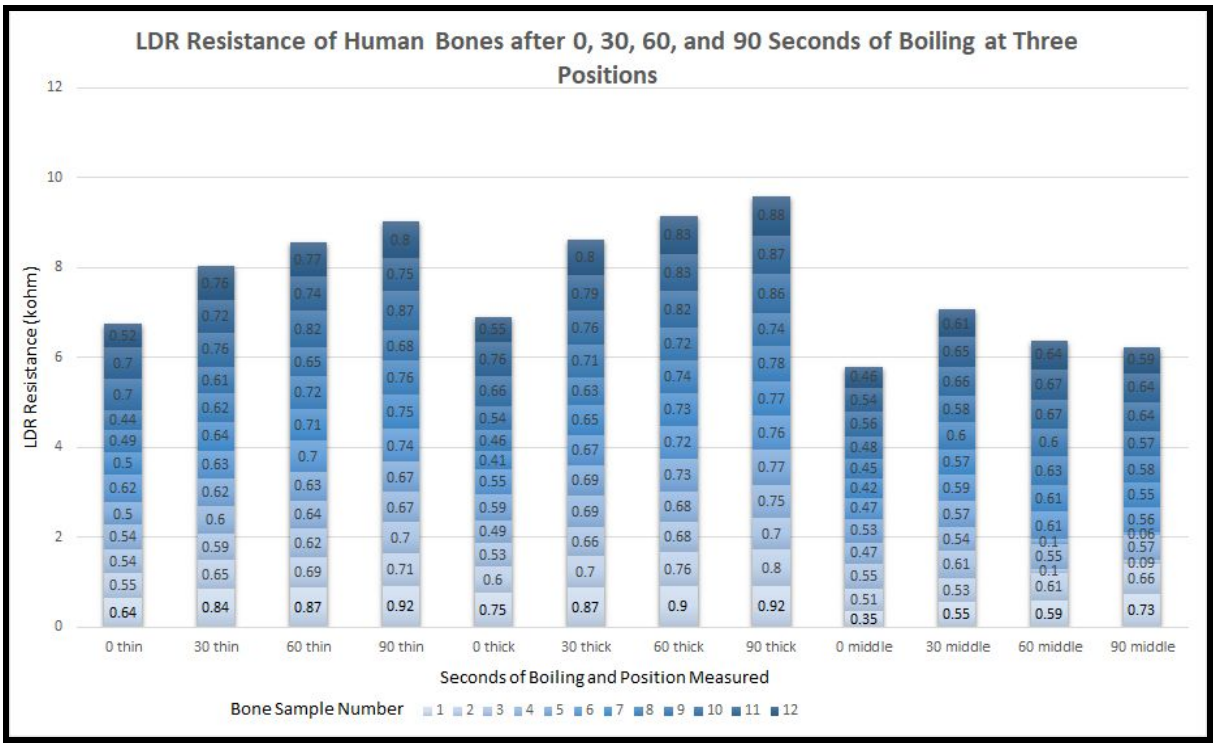


Graph 4: Aggregate Change in LDR Resistance (kohm) for all Thin (4.5 mm) Bone Samples

Keeping in mind the results of prior experimentation and the process of light microscopy, which uses extremely thin samples, it was determined that thin bone samples were best suited for testing. The average LDR resistance of the samples before boiling is 0.55 kohm, which is equivalent to approximately 20 lux. The average LDR resistance of the samples after boiling for one minute is 0.72 kohm, which is equivalent to approximately 8 lux. The average LDR resistance change, therefore, is 0.17 kohm, and the average decrease in light is 12 lux. There was an average increase of 30.9% in LDR Resistance and an average decrease of 60% in transmitted light.

Human Experimentation

For this experiment, a total of 12 cross sections were taken from the lower femur of a human 74 year old female. It is very likely that the bone was already osteoporotic. Each sample was measured in order to get baseline dimensions and classify the samples into one of two groups. One group, classified as “thin” was cut to an average thickness of 5mm. The other group, called “thick” was cut to approximately 10mm. There were 9 “thin” and 3 “thick” bone samples total. All boiling occurred at a temperature of 100°C, which was monitored using a digital cooking thermometer. LDR resistance was taken at the thinnest, thickest, and middle portions of the bone pre-boiling and after boiling for 30, 60, and 90 seconds.



Graph 5: LDR Resistance of Human Bones after 0, 30, 60, and 90 Seconds of Boiling at 100 degrees Celsius at Three Positions (Thinnest, Thickest, Middle) for Experimental Set 3

Samples #1-9 are "thin" samples and samples #10-12 are "thick". The greatest increase in LDR resistance occurred in the first thirty seconds of boiling. LDR Resistance continued to increase but at a lower pace, likely due to the lower amounts of proteins, fats, and minerals available to evacuate the bone during boiling at each interval. On the thin and thick positions, LDR continued to increase. On the middle position, however, LDR often decreased as whole pieces of spongy bone disintegrated during boiling. Human bones had different exact data points from bovine bone but showed a similar trend.

Discussion

The goals of this project were twofold: 1) determine if light can be used to detect differences in bone health, and 2) build a device that can determine a clinical score for bone porosity on a bone sample. To support this effort, post-mortem bovine bone was used as a substitute for human bone. A prototype was constructed that provides a self-contained system with a sample chamber/stage, a light source, a light dependent resistor, and the corresponding electronics to collect the data from the light sensor, and display them in an LCD to the operator of the system.

Twenty-two individual diaphyseal cross sections were taken from Texas Wagyu femur bones. Seven measurements were taken of each sample to determine a dimensional baseline of

the sample. The samples were subsequently subjected to boiling. One group of bones, samples 1-12, were boiled for 60 seconds. The second group, consisting of bones 13-22, were boiled boiled for three increments of 30 seconds each, with each sample being subjected to testing in the prototype after each boiling cycle. This experimental procedure is depicted in the Figure 4 flowchart.

In the initial calibration of the system, it was determined that higher levels of light register lower levels of resistance in the LDR. Therefore, higher levels of translucency will present as lower R values in the prototype. The converse is also true, so higher R values mean there is lower translucency in the bone. At this point, the primary measurement provided by the prototype is the resistance reported by the LDR. The T_e value shown is just a sample showing that calculations can be performed natively in the system, meeting the requirement that all analysis be self-contained in the system.

For the first experimental set, there were 12 diaphyseal cross sections of bovine bone. The first five were categorized as thin samples, and the remaining bone samples (seven) were categorized as thick samples. Thin samples have an average thickness of 4.5mm , and thick samples have an average thickness of 11.37mm . The average LDR resistance for thin samples before boiling was 0.47 kohm . The average LDR resistance for thin samples after boiling was 0.61 kohm . Therefore, there was an average increase of 29.8%. The average LDR resistance for thick samples before boiling was 0.74 kohm , and after boiling, the average LDR resistance was 0.83 kohm . Therefore, there was an average increase of 12.2%. This data was determined from Graph 2.

The second experimental set consisted of ten thin diaphyseal cross sections of bovine bone boiled at 100°C in three 30 second increments for a total of 90 seconds. LDR measurements were taken before boiling, after 30 seconds, 60 seconds, and 90 seconds of boiling. The average LDR resistance before boiling was 0.6 kohm . The average LDR resistance after 30 seconds of boiling was 0.76 kohm , meaning there was an average increase of 0.16 kohm. The average LDR resistance after 60 seconds total was 0.79 kohm . There was an average increase of 0.03 kohm from the 30 second sample. The average LDR resistance after 90 total seconds of boiling was 0.8 kohm . The average increase from the 60 second sample 0.01 kohm. Overall, there was an average increase of 0.2 kohm, or 33.3%. This data was determined from Graph 3.

Keeping in mind the results of the experimentation and the process of light microscopy, which uses very thin samples,[13] it was determined that the thin samples were best suited for testing. Therefore, the data in Graph 4 (the basis for this data), was all from thin samples. Because this data set consisted of all the thin samples across both experimental sets, a conversion to lux was deemed important. The average LDR resistance through healthy bone was 0.55 kohm (20 lux), while the average LDR resistance through porous bone was 0.72 kohm (8 lux). Therefore, there was an increase of 0.17 kohm and a decrease of 12 lux. There was a 30.9% reduction in LDR resistance and a 60% reduction in transmitted light. The lux data was interpolated from Graph 1.

The second phase of this project involved validating that the trends seen in bovine bone continued in human bone. For this experiment, a total of 12 cross sections were taken from the lower femur of a human 74 year old female. It is very likely that the bone was already osteoporotic. Each sample was measured in order to get baseline dimensions and classify the samples into one of two groups. One group, classified as "thin" was cut to an average thickness of 5mm. The other group, called "thick" was cut to approximately 10mm. There were 9 (#1-9) "thin" and 3 (#10-12) "thick" bone samples total. All boiling occurred at a temperature of 100°C, which was monitored using a digital cooking thermometer.

LDR resistance was taken at the thinnest, thickest, and middle portions of the bone pre-boiling and after 30, 60, and 90 seconds. The greatest increase in LDR resistance occurred in the first 30 seconds of boiling. LDR resistance continued to increase, but at a lower pace, likely due to the lower amounts of proteins, fats, and minerals available to evacuate the bone during boiling at each interval. On the thin and thick positions, LDR continued to increase. On the middle position, however, LDR often decreased as whole pieces of spongy bone disintegrated during boiling. Human bones had different exact data points from bovine bone but showed a similar trend. On the thin (5mm) bones, the thinnest section of the bone's LDR resistance increased by an average of 0.19 kohm. The thickest section of the bone experienced an average 0.23 kohm increase in LDR resistance. The middle section of the bone had a 0.02 kohm average increase in LDR resistance. On the thick (10 mm) bones, the thinnest section of the bone experienced an average increase of 0.17 kohm in LDR Resistance. The thickest bone section had an average LDR resistance increase of 0.22 kohm. The middle section's LDR resistance increased by an average of 0.1 kohm. This data is depicted in Graph 5.



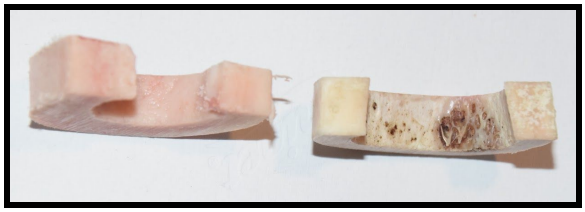


Figure 5 (Left): Severity of Osteoporosis Chart, courtesy of Endocrine Associates of Dallas

Figure 6 (Right): Images of Pre-Boiled Bone (Left) and Boiled Bone (Right), Displaying the Visual Changes of Boiling on Bone

The bones display differences in color, texture, size, visible porosity, and rigidity before and after boiling. Boiling the bone caused it to shrink and harden. The boiled bones, such as the one in Figure 6, show pores similar to clinical exemplar (Figure 5).

There are three main takeaways of this project. First, healthy bone samples were more translucent than porous bones, as indicated by lower LDR resistance values. Second, thin bone samples are better for clinical purposes. Third is that, while the data shows a correlation between porosity and translucence across samples in the same category as well as across categories, the results still do not provide a clinical analysis as to whether the bone that was tested is of a level of porosity that increases the risk of fracture. Normative samples of known porosities, if obtained, can be used to further validate the research.

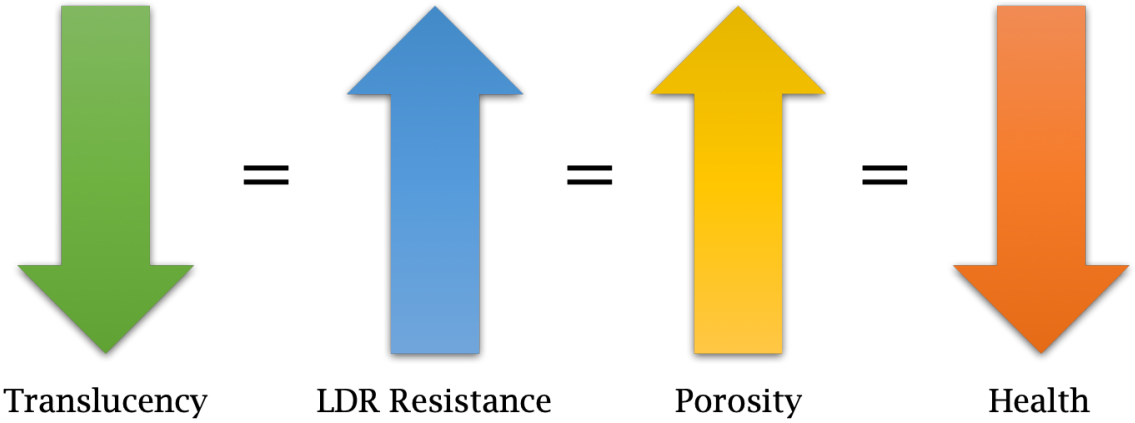


Figure 7: Project Takeaway- Lower translucency means higher LDR Resistance, higher porosity, and lower health (and Vice Versa)

Conclusions

To recap, there were three hypotheses for this project: 1) if a bone sample is exposed to a controlled source of light, then the amount of light transmitted through the bone will differ between healthy bone and porous bone, 2) if porosity increases in a bone sample, then translucency will decrease, and 3) human bone, when submitted to the same testing as bovine bone, will have different exact data points,but show a similar trend. All three hypotheses were supported by the results of the experiments, displayed in Graphs 2 , 3, 4, and 5.

Both graphs display that after boiling, samples display a higher LDR resistance, meaning they are less translucent and more porous. Graph 3 displays the increase in LDR resistance in 30 second increments for boiling. The most significant change occurred in the first 30 seconds of boiling. As boiling time increases, LDR resistance also continues to increase, although at a lower pace, which is likely due to the lower amounts of proteins remaining in the bone at each interval. These same findings were also evident in the human bone testing.

Post-mortem bone health can be analyzed through light resistance, as measuring light transmitted through a bone sample with an LDR is an effective means of determining differences in bone health. The LDR resistance values showed that as porosity increases, translucence

decreases. There is currently no clinical value to assign to the findings, which will be an instrumental aspect of this project's next steps.

Error Analysis

Although the results are quite consistent and appear conclusive, there are three main areas covering the possible sources of error.

Sample Placement: The conditions of the bone might not be consistent throughout, so testing different places on the bone before and after boiling will affect the results. Another aspect of sample placement is failure to completely cover the LED. Should that occur, there will be light leakage, which would give a false reading.

Voltage Fluctuation: If voltage fluctuates, then light intensity varies. Different intensities will affect the results. In this experiment, there were no visible indications of voltage fluctuation.

Unknown Porosity or Health of Initial Bone Sample: With an unknown porosity of the initial bone sample, it cannot be said whether or not the pre-boiled bone actually is a healthy bone. Therefore, what is being measured is technically relative porosity. This would affect a clinical calculation, which is essential to the completion of this project. Unknown initial bone porosity can also be applied to the fact that post-mortem bones are being used as healthy bones, which are likely actually not as healthy as healthy in-vivo bones.

Applications

All aspects of this project's applications at this point relate back to forensic analysis, which is important for health and medico-legal purposes.

Some examples of forensic analysis would include post-mortem identification of physical child abuse or starvation, in which identifying whether bone density is lower than average could provide evidence. Physical child abuse leads to hindered development, and not reaching peak bone mass while one is young leads to higher porosity in the future. Lack of calcium, mainly obtained through food, weakens bones, which could be determined using this test and indicate starvation.[3,9] This tool can also potentially be used to determine post-mortem age, an important factor in body identification, as bone porosity generally correlates with age. If the porosity indicates osteoporosis, comparison against records of osteoporosis diagnoses could help identification.[3,4,9]

The fact that this tool is simple to use and understand also makes it a valuable tool for students or less qualified scientists. Third-world countries, for example, often have a low supply of qualified scientists, and science receives nearly no funding, and so can use this low-cost, portable, versatile, and easy to understand device. This device is a good introduction to understanding bone and osteoporosis, making it applicable to students of forensics sciences.

Future Plans

While visible light is useful for forensic, post-mortem analysis, it is extremely unlikely that it can be used for in-vivo testing, as the wavelength will not penetrate human tissue. Other types of light, like UV and infrared, could be tested by replacing the visible light source and sensors with the applicable sources and sensors. There is a combination sensor that has UV, infrared, and visible light that might be a good option. It is also possible that, for these purposes, light is not optimal, and ultrasound is a better alternative.

An instrumental aspect of completing this project is to establish a clinical validation and/or conversion to T-score. Bones with a known porosity and health would be necessary for these purposes. Known porosity is necessary to validate the readings and results. Another aspect is to determine the mathematical formula and/or statistical analytics necessary to provide a final clinical diagnosis.

The ultimate goal of this project is to create a non-invasive, in-vivo bone density test for determining or monitoring osteoporosis. The device, unlike most other bone mineral density tests, will require little to no exposure to radiation. All of the plans above are geared towards achieving this goal.

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All images and other graphics were created by the student researcher unless stated otherwise.

Statement on Outside Assistance

Researcher's Name: Sophie Krajmalnik
Title of Paper: Determining Bone Health Utilizing Visible Light
What steps led you to formulate your hypothesis? I am an avid fan of Fox's *Bones*, the fictionalization of the life and work of forensic anthropologist Dr. Kathy Reichs. Through watching the show, the fact that bones are porous was drilled into me, and I was curious if light could be used to measure porosity of bones.

Where did you conduct the major part of your work? I conducted most of my work out of my home, but I worked in the ProPath Laboratories in order to work with human bones.
If you worked in an institutional setting, did you work on your project as part of a team or group? My project was individual, and I worked individually with Dr. Yan Lemeshev.
Describe what parts of the research you did on your own and what parts you received help. I did the majority of my work on my own. Enrique Krajmalnik, my mentor, acted as a sounding board and helped me work through my problems. Dr. Yan Lemeshev allowed me to use his facility and resources to test human bones and cut the bone according to my specifications.
Is your research current or a continuation of previous research? This research is current and not a continuation of previous research.